

Assignment Day – 5

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.
Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux



```
C:\Users\RAHUL>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=6ms TTL=64
Reply from 192.168.1.3: bytes=32 time=10ms TTL=64
Reply from 192.168.1.3: bytes=32 time=20ms TTL=64
Reply from 192.168.1.3: bytes=32 time=18ms TTL=64

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 20ms, Average = 13ms
```

```
2:18 PM >_
~ $ ping 192.168.1.101
PING 192.168.1.101 (192.168.1.101) 56(84) bytes of data.
64 bytes from 192.168.1.101: icmp_seq=1 ttl=128 time=2.21 ms
64 bytes from 192.168.1.101: icmp_seq=2 ttl=128 time=2.06 ms
64 bytes from 192.168.1.101: icmp_seq=3 ttl=128 time=1.89 ms
64 bytes from 192.168.1.101: icmp_seq=4 ttl=128 time=2.15 ms
64 bytes from 192.168.1.101: icmp_seq=5 ttl=128 time=1.94 ms

--- 192.168.1.101 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.891/2.050/2.212/0.108 ms
~ $
```

```
ESC / - HOME ↑ END PGUP
CTRL ALT ← ↓ → PGDN
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.



```
C:\Users\RAHUL>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482a:400::]
over a maximum of 30 hops:

  1    17 ms    58 ms    27 ms    2401:4900:8fc4:49ae:9a0a:4bff:fea0:79a1
  2     *        *        *        Request timed out.
  3    37 ms    223 ms   185 ms    fc00::8b
  4    98 ms    174 ms   258 ms    2404:a800:2a00:103::9
  5    95 ms     82 ms    78 ms    2404:a800::167
  6   150 ms    171 ms    62 ms    2001:4860:482a:400::
```

I run the tracert www.youtube.com command on windows device The route 6 hops to reach destination. The trace was completed successfully.

Number of Hops - 6

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.



```
C:\Users\RAHUL>ping 192.168.1.3  
  
Pinging 192.168.1.3 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Reply from 192.168.1.8: Destination host unreachable.  
Reply from 192.168.1.8: Destination host unreachable.
```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.



```
C:\Users\RAHUL>arp -a
```

```
Interface: 192.168.117.1 --- 0x3
Internet Address      Physical Address      Type
192.168.117.254       00-50-56-ef-41-7f     dynamic
192.168.117.255       ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
Interface: 192.168.136.1 --- 0xc
Internet Address      Physical Address      Type
192.168.136.254       00-50-56-f1-bc-4d     dynamic
192.168.136.255       ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
Interface: 192.168.1.8 --- 0x11
Internet Address      Physical Address      Type
192.168.1.1           98-0a-4b-a0-79-a1     dynamic
192.168.1.2           ca-fe-67-75-d4-3e     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

IP Address	MAC Address	Device / Gadget
192.168.1.1	98-0A-4B-A0-79-A1	Wi-Fi Router
192.168.1.2	CA-FE-67-75-D4-3E	Other connected device / Mobile
192.168.1.8	BC-03-58-7E-DF-74	My Laptop

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.



Network Setup

First, I connected my laptop and mobile phone to the same Wi-Fi network. My laptop was assigned the IP address 192.168.1.8, while the mobile device had an IP address in the same local network. The Wi-Fi router had the IP address 192.168.1.1.

Testing Connectivity Using Ping

I opened Command Prompt on the laptop and used the following command to test communication with the mobile phone:

ping 192.168.1.3

```
C:\Users\RAHUL>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=229ms TTL=64
Reply from 192.168.1.3: bytes=32 time=44ms TTL=64
Reply from 192.168.1.3: bytes=32 time=202ms TTL=64
Reply from 192.168.1.3: bytes=32 time=37ms TTL=64

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 37ms, Maximum = 229ms, Average = 128ms
```

Next, I disconnected the mobile phone from Wi-Fi and ran the ping command again. I received the following output:

```
C:\Users\RAHUL>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.1.8: Destination host unreachable.
Reply from 192.168.1.8: Destination host unreachable.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
```

Testing the Network Path Using Traceroute:

I used the Windows tracert command to check the route from my laptop to YouTube:

tracert www.youtube.com

```
C:\Users\RAHUL>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4828:400::]
over a maximum of 30 hops:

  1      3 ms      3 ms      3 ms  2401:4900:8fc4:49ae:9a0a:4bff:fea0:79a1
  2      *          *          *    Request timed out.
  3     80 ms     77 ms     76 ms  fc00::bb
  4     45 ms     39 ms     31 ms  2404:a800:2a00:103::9
  5     72 ms     71 ms     54 ms  2404:a800::167
  6    114 ms     58 ms    124 ms  2001:4860:4828:400::
```

```
C:\Users\RAHUL>arp -a
```

```
Interface: 192.168.117.1 --- 0x3
```

Internet Address	Physical Address	Type
192.168.117.254	00-50-56-ef-41-7f	dynamic
192.168.117.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
Interface: 192.168.136.1 --- 0xc
```

Internet Address	Physical Address	Type
192.168.136.254	00-50-56-f1-bc-4d	dynamic
192.168.136.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
Interface: 192.168.1.8 --- 0x11
```

Internet Address	Physical Address	Type
192.168.1.1	98-0a-4b-a0-79-a1	dynamic
192.168.1.2	ca-fe-67-75-d4-3e	dynamic
192.168.1.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

IP Address	MAC Address	Device / Gadget
192.168.1.1	98-0A-4B-A0-79-A1	Wi-Fi Router
192.168.1.2	CA-FE-67-75-D4-3E	Other connected device / Mobile
192.168.1.8	BC-03-58-7E-DF-74	My Laptop